

c) If A is an $m \times n$ matrix, then the range of the transformation $x \mapsto Ax$ is $\mathbb{R}^m$.

Solution: From part (b), we know that ~~each~~ the range of T is the set of all linear combinations of the columns of A . The columns of A has dimensions $m \times 1$. $\therefore$ The given statement is True. $\square$

d) Every linear transformation is a matrix transformation.

Solution: From the Note mentioned on Page 66, just below the definition of Linear Transformations, we know that the given statement is False. $\square$

e) A transformation T is linear if and only if $T(c_1v_1 + c_2v_2) = c_1T(v_1) + c_2T(v_2)$ for all v_1 and v_2 in the domain of T and for all scalars c_1 and c_2 .

Solution: Let, the transformation T be linear. The following conditions must hold: $T(v_1 + v_2) = T(v_1) + T(v_2)$, for all v_1 and v_2 in the domain of T
(Condition 1) $T(cv_1) = cT(v_1)$ for any arbitrary scalar c and for all v_1 in the domain of T .

$$T(c_1v_1 + c_2v_2) = T(c_1v_1) + T(c_2v_2) \text{ (Condition 1)} = c_1T(v_1) + c_2T(v_2) \text{ (Condition 2)}$$

$$\text{Let, } T(c_1v_1 + c_2v_2) = c_1T(v_1) + c_2T(v_2) \text{ for all } v_1 \text{ and } v_2 \text{ in the domain of } T$$

$$\text{and for all scalars } c_1 \text{ and } c_2.$$

$$\text{Let, } c_1 = c_2 = 1. \therefore T(1 \cdot v_1 + 1 \cdot v_2) = 1T(v_1) + 1T(v_2) \Rightarrow T(v_1 + v_2) = T(v_1) + T(v_2)$$

$\therefore$ Condition 1 satisfied.

$$\text{Let, } c_2 = 0 \therefore T(c_1v_1 + 0 \cdot v_2) = c_1T(v_1) + 0T(v_2) \Rightarrow T(c_1v_1) = c_1T(v_1)$$

$\therefore$ Condition 2 satisfied.

By defn, T is a linear transformation. $\therefore$ The given statement is True. $\square$

2) a) Every matrix transformation is a linear transformation.

Solution: On Page 66 under the section "Linear Transformation", it is mentioned that by Theorem 5 in Section 1.4, if A is $m \times n$, then the transformation $x \mapsto Ax$ has the properties $A(u+v) = Au + Av$ and $A(cu) = cAu$ for all u, v in $\mathbb{R}^n$ and all scalars c . $\therefore$ By defn, every matrix transformation is a linear transformation. $\therefore$ So the given statement is True. $\square$